Gojo 2020 pediatric ependymoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep on sample: sample iLISI 1.5→6.6 (θ=10), one connected component (0 islands); PF/ST/SP subgroups grade continuously across one cloud (subgroup iLISI stays ~1.6 — biology preserved).

Gate 2 — module geography → GO (non-branching)

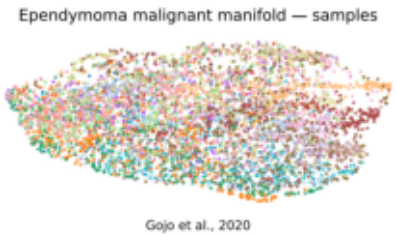
Non-branching: shared Gast/pSom body with an inverse terminal Crest gradient (Crest-high decile mean pseudotime = 0.97). Crest opposes Gast (r=−0.50) and pSom (r=−0.38); Gast~pSom weak (−0.20).

Gate 3 — cross-repression → NA

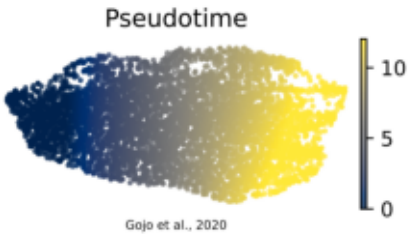
No competing Gast/pSom arms (weak −0.20, no bifurcation) — branching-only test does not apply.

STEP FIGURES

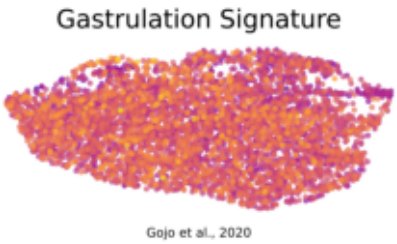
Manifold — samples



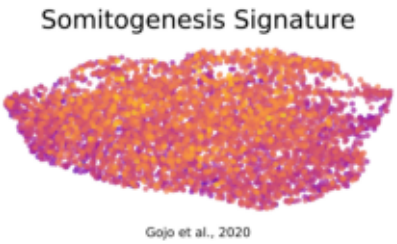
Pseudotime



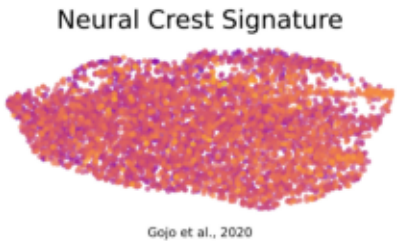
Gastrulation



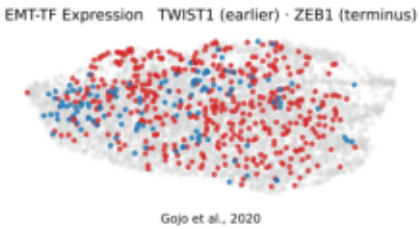
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric ependymoma (Gojo et al., 2020), SmartSeq2, 5,378 malignant cells across 20 samples (threshold ≥110/sample; PF/ST/SP molecular subgroups).

Verdict.
Pattern reproduces as the non-branching case — one continuous gastrulation/somitogenesis → neural-crest axis, no Gast/pSom bifurcation. Geometry: Linear-pSom/Crest. EMT-TFs overlay TWIST1 (earlier) ↔ ZEB1 (terminus).

Parameter choices

Min-cell threshold	≥110 / sample (largest gap 140→88, geometric mean)	**Cells retained**	5,551 → 5,378 (20/22 samples above threshold; 0 islands)
Overdispersed genes	4,997 (scFates find_overdispersed)	**Integration**	Harmony on sample θ=10 (n_pcs=20, spread=1.3); iLISI 6.6
Geometry	Non-branching; scFates curve N=30, λ=0.3, μ=100 (2 tips, 0 forks)	**Root**	Gast/pSom-high, Crest-low tip (Crest is late/terminal)
Module means (raw)	Gast +0.116 · pSom +0.329 · Crest +0.003 (pSom-dominant)	**EMT-TFs**	TWIST1 (earlier) ↔ ZEB1 (terminus); FOXC1/SNAI1/ZEB2/PRRX1 detected